

CS 215: Task 03: Due: 22-Aug Time: as mentioned in Piazza

- Please write (only if true) the honor code. If you used any source (person or thing) explicitly state it. This is an individual assignment.
 - Your answers should be in your handwriting.
 - Problems are not necessarily in order of difficulty.
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0. Please write the honor code in your handwriting, with the understanding that it is true.
1. A person tried by a 3-judge panel is declared guilty if at least 2 judges cast votes of guilty. Suppose that when the defendant is in fact guilty, each judge will independently vote guilty with probability 0.7, whereas when the defendant is in fact innocent, this probability drops to 0.2. If 70 percent of defendants are guilty, compute (see hints ahead) the conditional probability that judge number 3 votes guilty given that
- (a) judges 1 and 2 vote guilty,
- (b) judges 1 and 2 cast one guilty and one not guilty vote,
- (c) judges 1 and 2 both cast not guilty votes.

Hint: This problem has three elements.

- (a) Joint probabilities, the chain rule: $P(A, B, C) = P(A \mid B, C) P(B \mid C) P(C)$, and Bayes' rule
- (b) The second line with the word “independent” is the heart of the simplification of the joint probabilities
- (c) If you get stuck trying to solve this problem, consider changing the problem to a two judge scenario, with the artificial rule that a defendant is guilty if either of the two judges declares the defendant to be guilty.

2. Let X and Y be random variables with finite variances

$$\text{Var}(X) = \sigma_X^2 \quad \text{and} \quad \text{Var}(Y) = \sigma_Y^2.$$

- (a) As something unrelated to X and Y – but will be used later on in (d) – prove that if $\text{Var}(Z) = 0$, then

$$P\{Z = \mathbb{E}[Z]\} = 1.$$

- (b) Using the fact that variance is always nonnegative, show that

$$0 \leq \text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right)$$

implies that

$$-1 \leq \text{Corr}(X, Y).$$

(c) Similarly, use

$$0 \leq \text{Var} \left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y} \right)$$

to prove that

$$-1 \leq \text{Corr}(X, Y) \leq 1.$$

(d) Using the result from part (a), prove that if

$$\text{Corr}(X, Y) = 1$$

or

$$\text{Corr}(X, Y) = -1,$$

then X and Y must satisfy a linear relationship of the form

$$Y = a + bX,$$

where $b > 0$ when $\text{Corr}(X, Y) = 1$ and $b < 0$ when $\text{Corr}(X, Y) = -1$.

3. Consider n independent trials, each of which results in one of the three outcomes 1, 2, or 3, with corresponding probabilities p_1 , p_2 , and p_3 , where

$$p_1 + p_2 + p_3 = 1.$$

Let N_i denote the number of trials that result in outcome i .

Find

$$\mathbb{E}[N_i]$$

4. Let (X, Y, Z) have joint probability density

$$f_{X,Y,Z}(x, y, z) = \begin{cases} c(x + z), & 0 < x < y < z < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Determine the constant c .

(b) Compute

$$P(X + Y < Z).$$

Hint: Refer to example 4.3c from the textbook

(c) Compute

$$E[XZ].$$

5. Let $X \geq 0$ be a random variable satisfying

$$E[X] = 4, \quad E[X^2] = 24, \quad E[X^3] = 192.$$

Use Markov's inequality, applied appropriately to X , X^2 , and X^3 , to obtain a tight bound from the given information on

$$P(X \geq 10).$$

6. Let $X > 0$ be a random variable satisfying

$$E[X] = \mu.$$

For some constant $a > 0$, define

$$g(x) = x + \frac{a}{x}.$$

(a) Use Jensen's inequality to obtain a bound, in terms of μ and a , for

$$E\left[X + \frac{a}{X}\right].$$

- (b) If μ may also be chosen freely, determine the value of μ to minimize the Jensen bound.

7. Let $X_1, \dots, X_{25}$ be identically distributed random variables with

$$E[X_i] = 100, \quad \text{Var}(X_i) = 400.$$

Suppose that, for $i \neq j$,

$$\text{Cov}(X_i, X_j) = 40.$$

Let

$$\bar{X} = \frac{1}{25} \sum_{i=1}^{25} X_i.$$

- (a) Use Chebyshev's inequality to obtain a lower bound for

$$P(92 < \bar{X} < 108).$$

- (b) Compare this bound with the one obtained if the X_i 's were independent. Expand on your observations on the the two differing bounds.